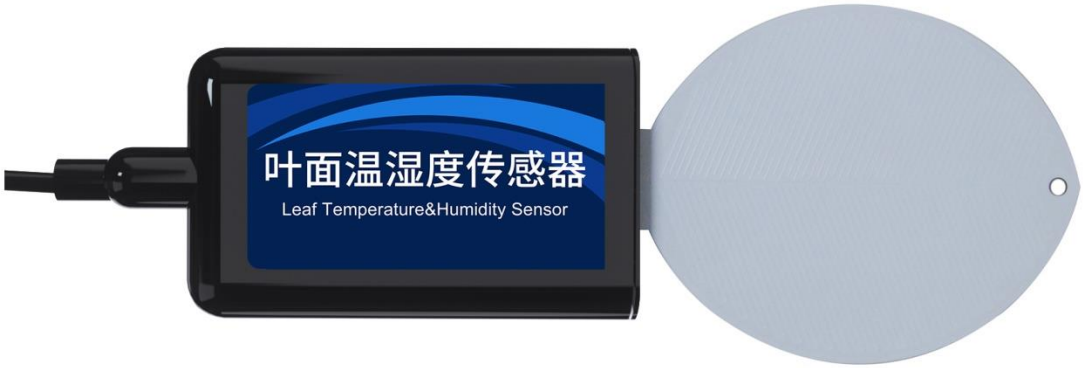
Leaf temperature and humidity sensor

Introduce



RD-LTH-O-01

1 Overview

The leaf temperature and humidity sensor is a composite sensor, which can accurately measure leaf temperature and humidity at the same time.

The temperature sensor is composed of high-precision NTC thermistor and transmitter. The transmitter has the functions of offset compensation and temperature calibration, which can accurately measure the temperature in the whole measuring range.

The humidity sensor is based on the principle of dielectric constant measurement, and simulates the leaf features by imitating the shape of blades. It can accurately measure the amount of water or ice by changing the dielectric constant of leaf surface. The sensitivity is good, and trace moisture or ice crystal residues on the leaf surface can be detected.

The leaf surface temperature and humidity sensor adopts waterproof design as a whole, which has low power consumption and can be monitored continuously for a long time. It is easy to install, and can be hung on the greenhouse or installed on the mast of the weather station.

2 Characteristic

1. Imitating leaf surface characteristics, temperature and humidity measurement is fast and accurate.
2. It has high precision temperature correction function and can output temperature signal independently.
3. The humidity measurement is sensitive and can accurately detect the moisture or ice crystal residue on the leaf surface.
4. Analog voltage/current can be selected, and RS485 signal can be output.
5. It is encapsulated with ABS and epoxy resin, which is waterproof and moisture-proof and has a longer service life.

3 Scope of application

This product can be widely used in environment, greenhouse, laboratory, breeding, warehousing, construction, high-grade buildings, industrial plants and other environmental temperature and humidity measurement of plant surface.

4 Product information

4.1 Technical Parameters

- Measurement parameters: leaf surface temperature and humidity

- Measuring range:-30 ~ 70 ℃ (0 ~ 50 ℃ or other arbitrary measuring range can be customized); 0-100% RH
- Resolution: 0.1 ℃; 0.1%
- Measurement accuracy: 0.2 ℃ (@ 25 ℃); ± 5% RH
- Output signal:

A: Voltage signal (0 ~ 2V, 0 ~ 5V, 0 ~ 10V)

B: 4 ~ 20mA (current loop)

C: RS485 (standard Modbus-RTU protocol, device default address: 01)

- Power supply voltage: 5 ~ 24V (when the output signal is 0 ~ 2V and RS485)

12 ~ 24V (when the output signal is 0 ~ 5V, 0 ~ 10V and 4 ~ 20mA)

- Output impedance: 10K Ω (voltage output type)
- Response time: ≤ 2S
- Working range: temperature:-30 ~ 70 ℃; Humidity: 0 ~ 100% RH
- Annual humidity stability: ± 1% RH
- Humidity nonlinearity: ± 1% RH

4.2 Physical Parameters

Leaf size: 76mm*61mm

Shell material: ABS

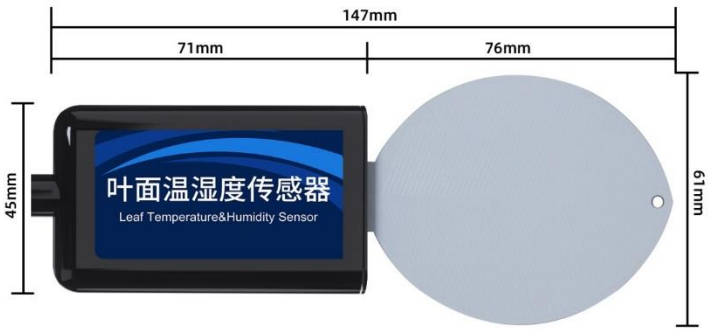
Sealing material: Epoxy resin

Cable specification: 2 meters as standard (other cable lengths can be customized, up to 1200 meters)

4.3 Impedance Requirements for Current Signals

Supply voltage	9V	12V	20V	24V
Maximum impedance	125 Ω	250 Ω	500 Ω	> 500 Ω

5 Shape specification



6 Usage

The leaf surface temperature and humidity sensor can be connected with various data collectors, data acquisition cards, remote data acquisition modules and other equipment with differential input. The wiring description is as follows:

1. For the RS485 output

Wire colour	Definition
Red wire	Positive power supply
Black	Negative power supply
Yellow	RS485A
Green	RS485B

2. For the analog voltage and current output

Wire colour	Definition
Red wire	Positive power supply
Black	Negative power supply
Green	Humidity analog voltage and current output
Yellow	Temperature analog voltage and current output

7 Data conversion method

7.1. Analog signal output

T: Temperature, unit: ℃;

H: Humidity in% RH;

V: Voltage value collected by collector, unit: V;

A: The current value collected by the collector, unit: mA;

Output signal	Leaf surface temperature conversion method			Conversion method of foliar humidity
	-30 ~ 70 ℃	0 ~ 50 ℃	-20 ~ 80 ℃	0-100% RH
0 ~ 2V DC	T=50*V-30	T=25*V	T=50*V-20	H=50*V
0 ~ 5V DC	T=20*V-30	T=10*V	T=20*V-20	H=20*V
0 ~ 10V DC	T=10*V-30	T=5*V	T=10*V-20	H=10*V
4 ~ 20mA	T=6.25*A-55	T=3. 125*A-12. 5	T=6.25*A-45	H=6.25*A-25

7.2 RS485 Signal (Default Address 01):

Standard Modbus-RTU protocol, baud rate: 9600; Check bit: None; Data bits: 8; Stop bit: 1

1. Address Modification

For example, the sensor with address 1 is changed to address 2, and the host → slave

Original address	Function code	Start register high	Start register low	High starting address	Low starting address	Low CRC16	CRC16 High
0X01	0X06	0X00	0X30	0X00	0X02	0X08	0X04

If the sensor receives correctly, the data returns according to the original route.

Remarks: If you forget the original address of the sensor, you can use the broadcast address 0XFE instead. When using 0XFE, the host can only connect with one slave, and the return address is still the original address, which can be used as an address query method.

2. Query Data

Inquire the data (leaf surface temperature, humidity) of the sensor (address 1), host → slave

Address	Function code	Start register address high	Start register address low	High register length	Low register length	Low CRC16	CRC16 High
0X01	0X03	0X00	0X00	0X00	0X02	0XC4	0X0B

If the sensor receives correctly, return the following data, slave → host

Address	Function code	Data length	Register 0 High data	Register 0 Low data	Register 1 data high	Register 1 data low	Low CRC16	CRC16 High
0X01	0X03	0X04	0XFF	0XDD	0X01	0X00	0X5B	0X8D
			Leaf surface temperature:-3.5 ℃ Negative temperature is expressed by complement code		Leaf humidity: 25.6% RH			

8 Matters needing attention in use

1. When you receive the product, please check whether the package is in good condition, and check whether the sensor model and specification are consistent with the product you purchased.

2. The installation place should be far away from chemical corrosion environment.

3. Sensors and wires should be far away from high-voltage electricity and heat sources.

4. Avoid scratching of the instrument, maintain the integrity of the external protective film, and prolong the service life of the instrument.

5. Rough treatment of the instrument will destroy the internal circuit board and precise structure.

9 Product warranty

The warranty period of this product is one year. From the date of delivery, within 12 months, due to sensor quality problems (non-man-made damage) caused by the failure, the company is responsible for free repair or replacement, beyond the warranty period only charge cost.